4. $\text{lift}(X \to Y) = \dfrac{c(X \to Y)}{s(Y)}$

$$s(Y) = \frac{\sigma(Y)}{N} = P(Y)$$

$$\text{lift}(X \to Y) = \frac{c(X \to Y)}{P(Y)}$$

$$c(X \to Y) = \frac{s(X \cap Y)}{s(X)} = \frac{P(X \cap Y)}{P(X)} = P(Y \mid X)$$

If lift $= 1$,

$$P(Y \mid X) = P(Y)$$

This implies Y is independent of X.

[$P(X \cap Y) = P(X)\,P(Y)$ is classic condition of independent event]

5. a)

TID	Items
1	bananas, carrots, figs
2	apples, donuts,
3	bananas, carrots, donuts, figs
4	carrots, eggs
5	apples, eggs
6	donuts
7	apples, bananas, carrots
8	apples, carrots, eggs
9	carrots, donuts, eggs, figs
10	apples, donuts, figs

Our unique items = apples, bananas, carrots, donuts, eggs, figs

N = 10. support threshold = 0.3

$$s(\text{apples}) = \frac{\sigma(\text{apples})}{N} = \frac{5}{10} = 0.5 > 0.3 \checkmark$$

$$s(\text{bananas}) = \frac{\sigma(\text{bananas})}{N} = \frac{3}{10} = 0.3 = 0.3 \checkmark$$

$$s(\text{carrots}) = \frac{\sigma(\text{carrots})}{N} = \frac{5}{10} = 0.5 > 0.3 \checkmark$$

$$s(\text{donuts}) = \frac{\sigma(\text{donuts})}{N} = \frac{5}{10} = 0.5 > 0.3 \checkmark$$

$$s(\text{eggs}) = \frac{\sigma(\text{eggs})}{N} = \frac{4}{10} = 0.4 > 0.3 \checkmark$$

$$s(\text{figs}) = \frac{\sigma(\text{figs})}{N} = \frac{4}{10} = 0.4 \geq 0.3 \checkmark$$

Hence for set size 1, frequency sets :-
{ apples }, { bananas }, { carrots }, { donuts },
{ eggs }, { figs }

For set size 2 :-

$$s(\text{apples, bananas}) = \frac{\sigma(\text{apples} \cap \text{Bananas})}{10} = \frac{1}{10} = 0.1 < 0.3 \times$$

$$s(\text{apples, carrots}) = \frac{\sigma(\text{apples} \cap \text{carrots})}{10} = \frac{2}{10} = 0.2 < 0.3 \times$$

$$s(\text{apples, donuts}) = \frac{\sigma(\text{apples} \cap \text{donuts})}{10} = \frac{2}{10} = 0.2 < 0.3 \times$$

$$s(\text{apples, eggs}) = \frac{\sigma(\text{apples} \cap \text{eggs})}{10} = \frac{2}{10} = 0.2 < 0.3 \times$$

$$s(\text{apples, figs}) = \frac{\sigma(\text{appls} \cap \text{figs})}{10} = \frac{1}{10} = 0.1 < 0.3 \times$$

$$s(\text{bananas, carrots}) = \frac{\sigma(\text{bananas} \cap \text{carrots})}{10} = \frac{3}{10} = 0.3 = 0.3 \checkmark$$

$$s(\text{bananas, donuts}) = \frac{\sigma(\text{bananas} \cap \text{donuts})}{10} = \frac{1}{10} = 0.1 < 0.3 \times$$

$$s(\text{bananas, eggs}) = \frac{\sigma(\text{bananas} \cap \text{eggs})}{10} = \frac{0}{10} = 0 < 0.3 \times$$

$$s(\text{bananas, figs}) = \frac{\sigma(\text{bananas} \cap \text{figs})}{10} = \frac{2}{10} = 0.2 < 0.3 \times$$

$$s(\text{carrots}, \text{donuts}) = \frac{\sigma(\text{carrots} \cap \text{donuts})}{10} = \frac{2}{10} = 0.2 < 0.3 \times$$

$$s(\text{carrots}, \text{eggs}) = \frac{\sigma(\text{carrots} \cap \text{eggs})}{10} = \frac{3}{10} = 0.3 = 0.3 \checkmark$$

$$s(\text{carrots}, \text{figs}) = \frac{\sigma(\text{carrots} \cap \text{figs})}{10} = \frac{3}{10} = 0.3 = 0.3 \checkmark$$

$$s(\text{donuts}, \text{eggs}) = \frac{\sigma(\text{donuts} \cap \text{eggs})}{10} = \frac{1}{10} = 0.1 < 0.3 \times$$

$$s(\text{donuts}, \text{figs}) = \frac{\sigma(\text{donuts} \cap \text{figs})}{10} = \frac{3}{10} = 0.3 = 0.3 \checkmark$$

$$s(\text{eggs}, \text{figs}) = \frac{\sigma(\text{eggs} \cap \text{figs})}{10} = \frac{1}{10} = 0.1 < 0.3 \times$$

2 itemset are :-
{bananas, carrots}, {carrots, eggs},
{carrots, figs}, {donuts, figs}

2 form 3-itemset, we use $F_{k-1} \times F_{k-1}$ method.
Using which we get :-
{carrots, eggs, figs}

[Thes were sorted lexographically & we join {carrots, eggs} with {carrots, figs} since carrot is same]

However, in {carrots, eggs, figs}, subset {eggs, figs} is not frequent. Hence no 3-item frequent set is present.

b) Our frequency sets :-
{apples}, {bananas}, {carrots}, {donuts},
{eggs}, {figs}, {bananas, carrots},
{carrots, eggs}, {carrots, figs}, {donuts, figs}

Since {apples} is not part of any 2-item frequent set, {apples} is a maximal frequent set.

All 2 item frequent sets are maximal frequent sets since we don't have any 3-item frequent sets.

∴ Maximal frequent sets :- {apples},
{bananas, carrots}, {carrots, eggs},
{carrots, figs}, {donuts, figs}.

c) support = 0.3 confidence = 0.7

Using {bananas, carrots}

$s(\text{bananas} \rightarrow \text{carrots}) = 0.3 \geq 0.3$

$c(\text{bananas} \rightarrow \text{carrots}) = \dfrac{s(\text{bananas} \cup \text{carrots})}{s(\text{bananas})} = \dfrac{0.3}{0.3} = 1 \geq 0.7$

Hence {bananas → carrots} is an association rule.

6. a) Minimum support = 0.2   Min Count = 2

TID	Items
1	{apples}
2	{apples, carrots}
3	{apples, carrots, donuts}
4	{apples, bananas, eggs}
5	{apples, bananas, carrots, donuts}
6	{bananas, donuts, eggs}
7	{bananas, carrots, figs}
8	{apples, bananas, carrots}
9	{apples, bananas, donuts, eggs}
10	{apples, carrots, eggs}

σ(apples) = 8 σ(bananas) = 6

σ(carrots) = 6 σ(donuts) = 4

σ(eggs) = 4 σ(figs) = 1

We sort them in order:-

apples > bananas > carrots > donuts > eggs

& we drop figs since its count < 2.

TID	items
1	{apples}
2	{apples, carrots}
3	{apples, carrots, donuts}
4	{apples, bananas, eggs}
5	{apples, bananas, carrots, donuts}
6	{bananas, donuts, eggs}
7	{bananas, carrots, }
8	{apples, bananas, carrots}
9	{apples, bananas, donuts, eggs}
10	{apples, carrots, eggs}

Building FP tree :-

ϕ

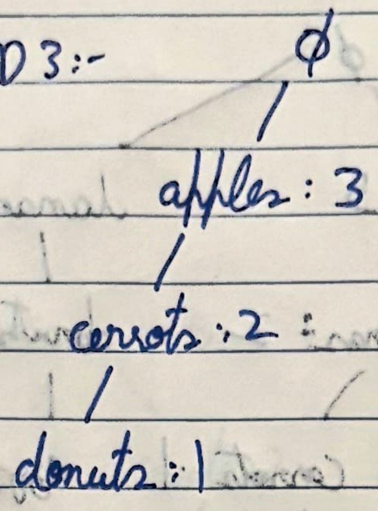
After TID 1 :-

ϕ
|
apples : 1

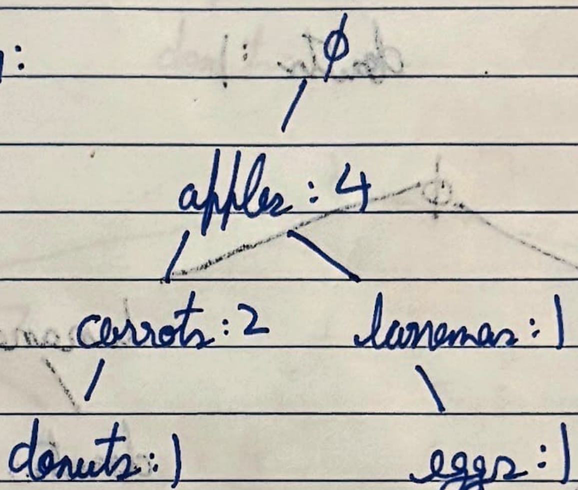
After TID 2 :-

ϕ
|
apples : 2
|
carrots : 1

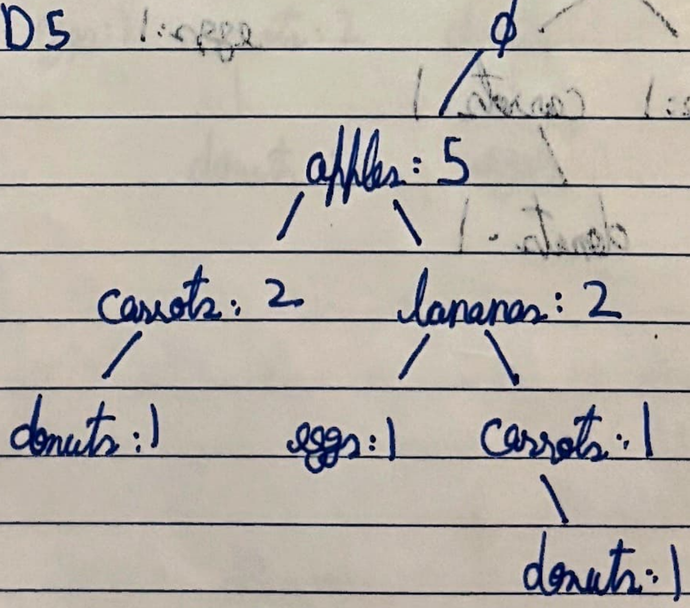
After TID 3:-



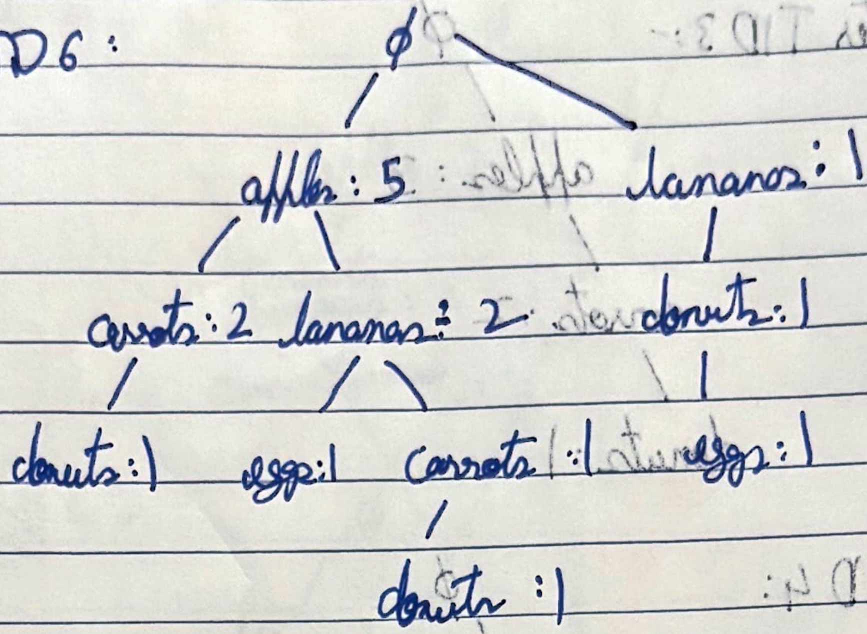
TID 4:



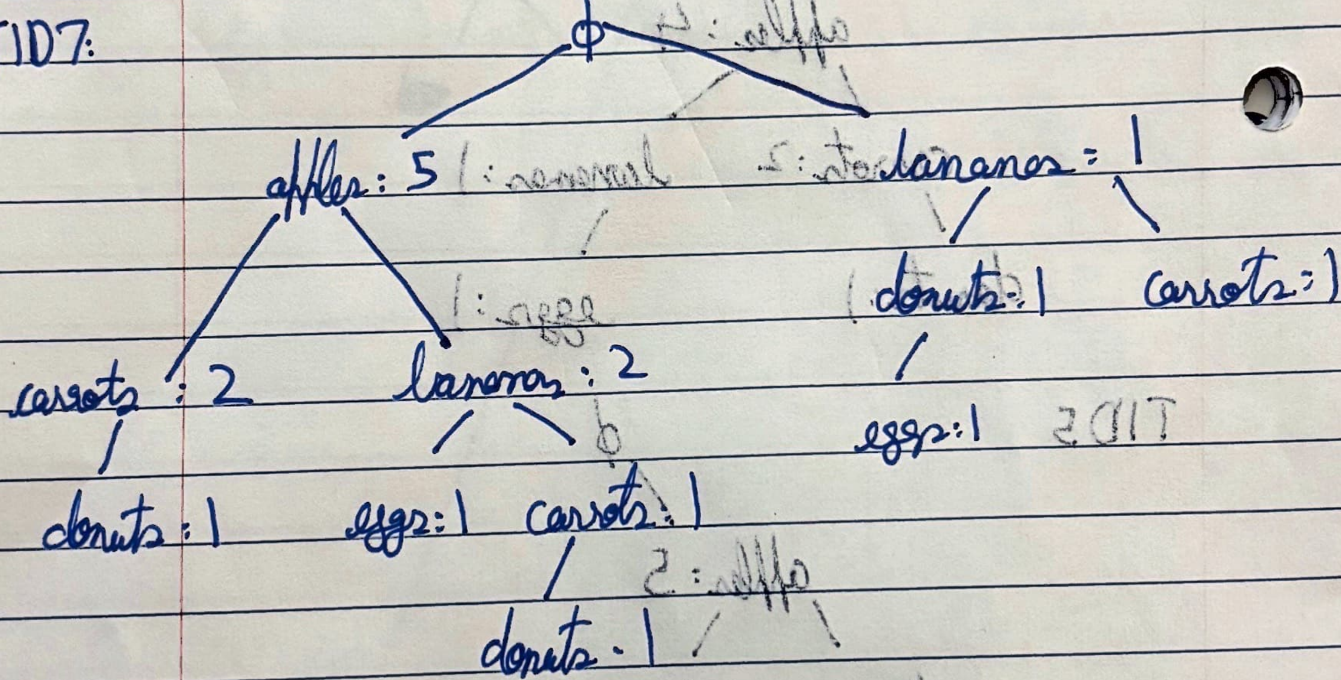
TID 5



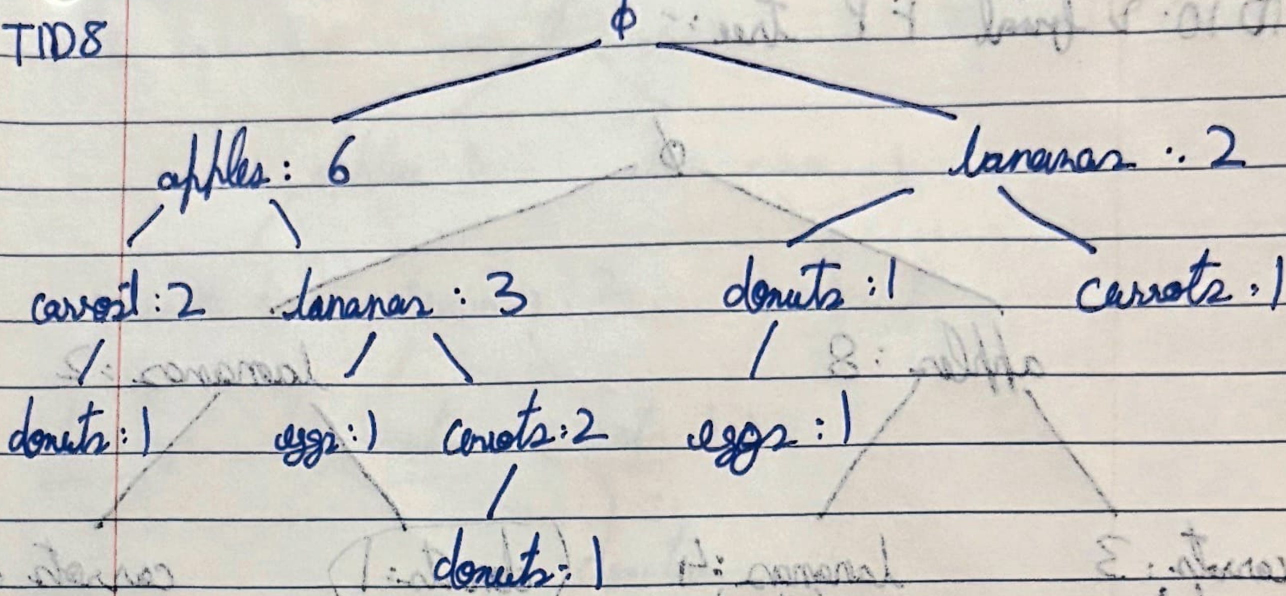
TID6 :


ϕ
apples : 5
bananas : 1
carrots : 2
bananas : 2
donuts : 1
donuts : 1
eggs : 1
carrots : 1
eggs : 1
donuts : 1

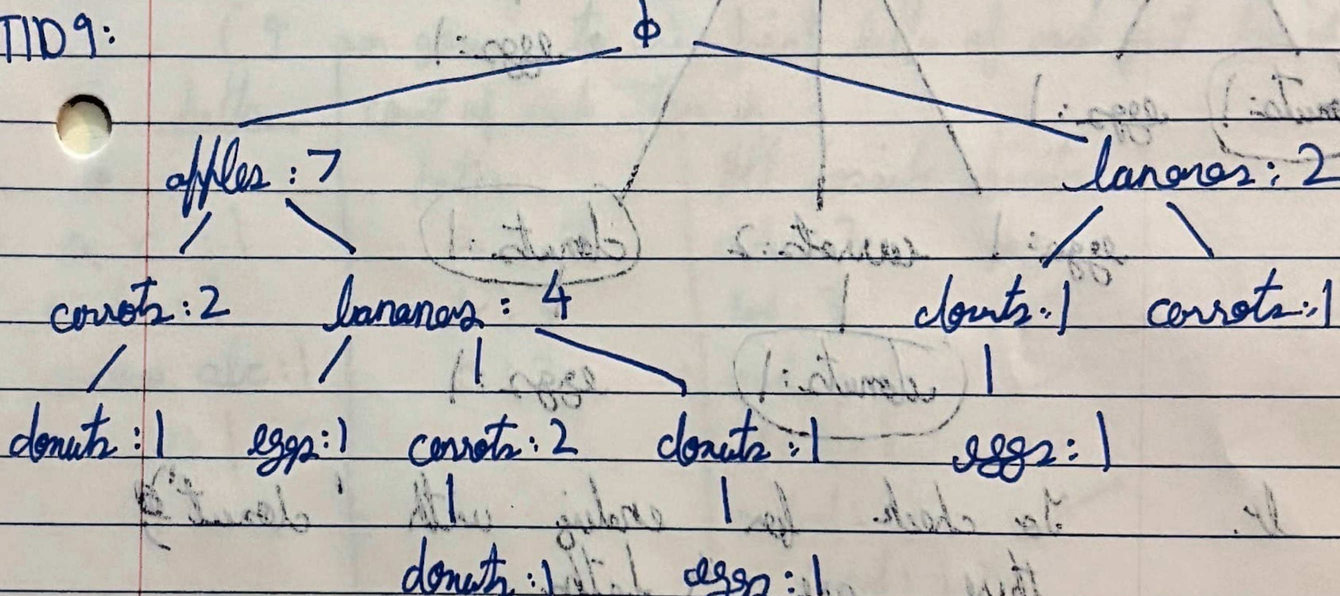
TID7 :


ϕ
apples : 5
bananas : 1
donuts : 1
carrots : 1
eggs : 1
carrots : 2
bananas : 2
donuts : 1
eggs : 1
carrots : 1
donuts : 1

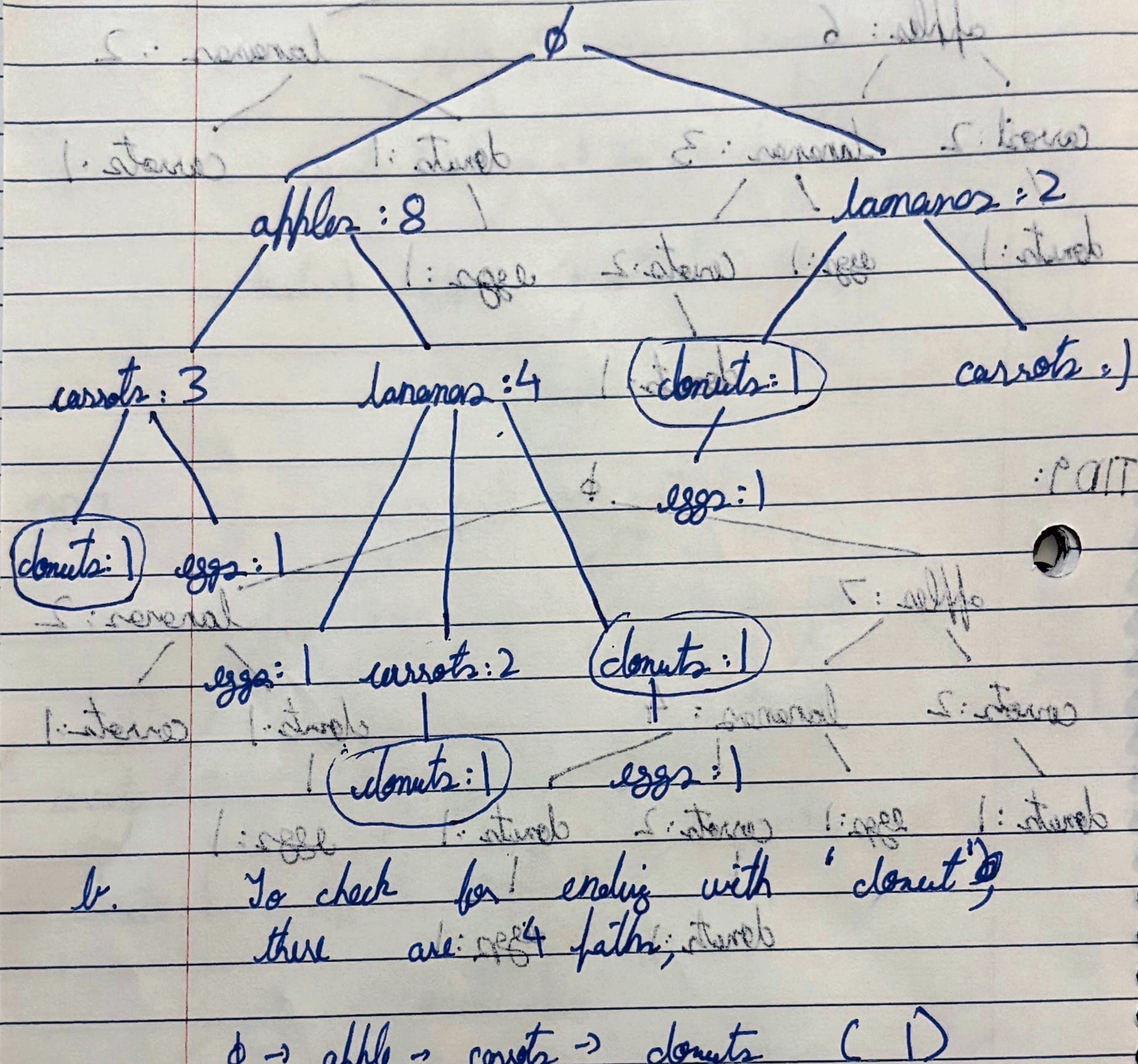
TID8


ϕ
apples: 6
bananas: 2
carrots: 2
bananas: 3
donuts: 1
carrots: 1
donuts: 1
eggs: 1
carrots: 2
eggs: 1
donuts: 1

TID9:

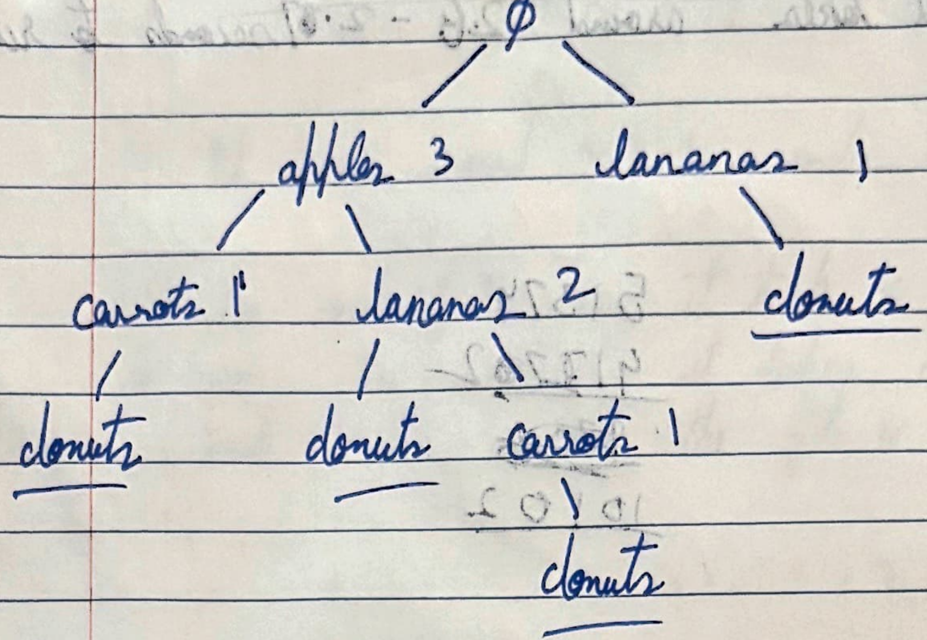

ϕ
apples: 7
bananas: 2
carrots: 2
bananas: 4
donuts: 1
carrots: 1
donuts: 1
eggs: 1
carrots: 2
donuts: 1
eggs: 1
donuts: 1
eggs: 1

TID10 & final FP tree :-



b. To check for ending with 'donut',
there are 4 paths;

ϕ → apple → carrots → donuts (1)
ϕ → apples → bananas → donuts (1)
ϕ → apples → bananas → carrots → donuts (1)
~~ϕ → apples~~ → ~~b~~
ϕ → bananas → donuts (1)



(I am going to use first letter of each fruit for it)

paths	count of each item in paths	All possible itemsets
ac : 1	a : 3	ad : 3 ✓
ab : 1	b : 3	bd : 3 ✓
~~abc~~ abc : 1	c : 2	cd : 2 ✓
b : 1		acd : 2 ✓
		abd : 2 ✓
		bcd : 1 ✗
		abcd : 1 ✗

~~Hence~~ Hence, frequency itemsets ending with donut are :-
{apple, donut}, {banana, donut}, ~~{carrot, donut}~~
{carrots, donut}, {apple, carrot, donut},
{apple, banana, donut}